SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
ASSESSMENT TOOL 1 – ANALYTICAL PROBLEM SOLVING

Course Code:	CSA06	**Course Title:**	Design and Analysis of Algorithms
CO Assessed:	CO2 – Manipulation (BL1, BL2, BL3)	**Assessment:**	Assessment Tool 1 – Analytical Problem Solving
Weightage:	40% of CIA	**Total Marks:**	100
CO2 Statement:	*Apply Brute Force technique to solve sorting, searching, Closest-Pair and Convex-Hull problems (BL1, BL2, BL3)*		
Student Name:	M Vishnu Vardhan	**Reg. No.:**	192472087
Section / Batch:	2024	**Date:**	27/07/26

Instructions to Students
- Answer the following question. Total Marks: 100.
- Analyze each problem carefully before applying the appropriate Brute Force technique.
- Clearly show all intermediate steps, comparisons, iterations, and logical reasoning used in solving the problems.
- Apply suitable algorithms for sorting, searching, Closest-Pair, and Convex-Hull problems wherever required.
- Justify the correctness and efficiency of the proposed solution using appropriate complexity analysis.
- Use diagrams, tables, or stepwise illustrations wherever necessary for better explanation.
- Maintain neat presentation, proper algorithmic notation, and structured analytical reasoning throughout the answers.

Question Type	Focus Area	Questions
Application-Based Questions	Apply Brute Force techniques for sorting, searching, Closest-Pair and Convex-Hull problem analysis	Q1

SECTION A – APPLICATION BASED QUESTIONS

Q45. Linear Search – Negative Number Detection Given the unsorted dataset [12, -7, 25, 39, -14, 48, 56], apply Linear Search to find the element -14. Clearly interpret the problem and explain the search process step-by-step. Count the number of comparisons required before locating the target. Analyze the best-case, average-case, and worst-case time complexity. Verify the correctness of the result. Interpret the efficiency of Linear Search when searching for negative values in an unsorted list.

Code of Conduct
I N. Thanveer Basha (192372122) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: N. Thanveer Basha___________

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20%	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30%	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20%	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20%	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10%	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Marks Evaluation:

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20%				
Application of Concepts	30%				
Problem-Solving Approach	20%				
Accuracy of Solution	20%				
Clarity	10%				
Total Marks (100):					

Faculty In-charge	Course Coordinator	Head of Department
Name:	Name:	Name:
Signature: _______________	Signature: _______________	Signature: _______________
Date: ___________________	Date: ___________________	Date: ___________________

09. Brute Force Maximum Subarray Problem

Given an array [-2, 3, 4, -1, -2, 1, 5, -3], apply the brute force approach to find the maximum subarray sum. Clearly interpret the problem and explain the method used to examine all subarrays. Show the calculations for different subarrays. Count the number of comparisons made. Identify the maximum sum and verify the result. Analyze the time complexity. Interpret the efficiency of this approach

9:- Brute Force Maximum Subarray

Given an array [-2, 3, 4, -1, -2, 1, 5, -3]

A = [-2, 3, 4, -1, -2, 1, 5, -3]

Steps:-

1. Select a starting index.
2. Index extend the subarray one element at a time.
3. calculate the sum of each subarray.
4. compare it with the current maximum sum.

An array size of n = 8

$$\frac{n(n+1)}{2} = \frac{8(9)}{2}$$

$$= \frac{\cancel{72}^{36}}{\cancel{2}_1}$$

$$= 36$$

Therefore, 36 subarrays are examined.

Number of comparisons:-

36 subarrays is compared with the current maximum sum.

Maximum sum:-

The heighest sum obtained is

$\boxed{10}$

[3, 4, -1, -2, 1, 5]

Verification

$3+4-1-2+1+5=10$

Time complenity:-

$T(n) = O(n^2)$

$O(n^3)$

Space complenity:-

(sum & MaxSum)

$\boxed{O(1)}$

calculations of Different subarrays

Index 0

subarray	sum
[-2]	-2
[-2,3]	1
[-2,3,4]	5
[-2,3,4,-1]	4
[-2,3,4,-1,-2]	2
[-2,3,4,-1,-2,1]	3
[-2,3,4,-1,-2,1,5]	8
[-2,3,4,-1,-2,1,5,-3]	5

Index 1

Subarray	sum
[3]	3
[3,4]	7
[3,4,-1]	6
[3,4,-1,-2]	4

10:- Nearest Neighbor Distance (20 points)

Given points

$A(1,2)$, $B(4,6)$, $C(7,8)$, $D(2,3)$, $E(3,5)$

Distance formula:-

$$d = \sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$$

compute All pair Distances:-

pair	Distance
A-B	$\sqrt{25}=5$
A-C	$\sqrt{72}=8.49$
A-D	$\sqrt{2}=1.41$
A-E	$\sqrt{25}=5$
B-C	$\sqrt{13}=3.61$
B-D	$\sqrt{13}=3.61$
B-E	$\sqrt{2}=1.41$
C-D	$\sqrt{50}=7.07$
C-E	$\sqrt{13}=3.61$
D-E	$\sqrt{13}=3.61$

Num. of comparisons

$n = 5$ points

$$\frac{n(n-1)}{2} = \frac{5 \times 4}{2} = 10$$

Total comparisons = 10

Nearest pair

Minimum distance

$$\boxed{\sqrt{2} \approx 1.41}$$

A(1,2) – D(2,3)
B(4,6) – E(5,5)

Time Complexity:-

Best case $O(n^2)$
Average case $O(n2)$
Worst case $O(n^2)$

Nearest pair A-D and B-E

Minimum Distance $\sqrt{2} \approx 1.41$

Total comparison 10

Time complexity $O(n^2)$